

Cage Allocation and Batch-Mixing Report

How do we fill finishing cages while keeping meaningful comparisons?

19	16	3	12.5 kg/m3	0
active cages	occupied	available (1B, 1C, 2E)	highest density (1E)	batch combinations on record

Current occupancy and density (23 Sep)

Cage	Stage	Volume m3	Batch	Origin	Fish	ABW g	Biomass kg	kg/m3	Transfer readiness
1A	nursing	125	03.26aK	Kipili	7,690	47.2	363	2.9	No
1D	nursing	125	03.26c	Kimbwela	16,173	96.0	1,553	12.4	No
1E	nursing	125	03.26aK	Kipili	28,707	54.5	1,565	12.5	No
1F	nursing	125	08.26	Kimbwela	19,803	6.5	129	1.0	No
2A	nursing	125	09.26a	Kimbwela	5,612	5.4	30	0.2	No
2B	nursing	125	05.26b	Kimbwela	12,996	41.7	542	4.3	No
2C	grow-out	125	02.26b	Kimbwela	9,582	132.1	1,266	10.1	Approaching
2D	grow-out	125	02.26b	Kimbwela	4,389	222.1	975	7.8	Ready (>150 g)
3A	nursing	125	05.26aK	Kimbwela	25,409	47.0	1,194	9.6	No
3B	nursing	125	05.26cK	Kipili	29,796	34.3	1,022	8.2	No
3C	nursing	125	06.26a	Kimbwela	41,104	22.1	908	7.3	No
3D	nursing	125	03.26b	Kimbwela	2,964	116.4	345	2.8	Approaching
3E	nursing	125	07.26a	Kimbwela	56,839	11.5	654	5.2	No
3F	nursing	125	07.26b	Kimbwela	31,033	10.5	326	2.6	No
G1.A	grow-out	750	02.26cK	Kipili	61,938	63.0	3,902	5.2	No
G1.D	grow-out	790	02.26aK	Kipili	37,163	82.9	3,081	3.9	No
1B	nursing	125	-	-	0	-	0	0.0	Available
1C	nursing	125	-	-	0	-	0	0.0	Available
2E	grow-out	125	-	-	0	-	0	0.0	Available

Usable volume = recorded system volume; net-pen depth and oxygen limits are not applied. Density is biomass / volume. A farm density ceiling is not recorded, so no cage is flagged as overstocked; the plan's own limit is not defined.

Proposed combinations (similar ABW, same stage, compatible origin first)

Cages (ABW, fish)	Batches	Origins	Combined load	Comment
1D (96.0 g, 16,173) + 3D (116.4 g, 2,964)	03.26c + 03.26b	Kimbwela + Kimbwela	19,137 fish / 1,898 kg in one 125 m3 cage = 15.2 kg/m3	Frees 1 cage; both are 96-116 g. 3D is already being harvested at 110-134 g.
1A (47.2 g, 7,690) + 2B (41.7 g, 12,996)	03.26aK + 05.26b	Kipili + Kimbwela	20,686 fish / 905 kg in one 125 m3 cage = 7.2 kg/m3	Frees 1 cage; mixes origins - keep both batch IDs with fish counts so comparisons remain possible.
3A (47.0 g, 25,409) + 1E (54.5 g, 28,707)	05.26aK + 03.26aK	Kimbwela + Kipili	54,116 fish / 2,759 kg in one 125 m3 cage = 22.1 kg/m3	Too dense for a 125 m3 cage; only workable in G1.D / G1.A volume.
2C (132 g, 9,582) + 2D (222 g, 4,389)	02.26b + 02.26b	Same batch	13,971 fish / 2,241 kg in one 125 m3 cage = 17.9 kg/m3	ABW differs by 68%; regrade first, do not combine as-is.

Preserving batch contribution: the database has batch_combination and batch_combination_component tables, with a derived batch and per-component fish counts (none created yet). Combine through that route (not by re-labelling) so each source batch's share survives for performance comparison. Mixed cages should be sampled by origin before the next monthly close.

Cycle records that no longer match cages

Cage	Cycle batch still 'ongoing'	Where the fish are now	Issue
1B	02.26cK (cycle 13)	G1.A	Cage is 'available' but cycle open
1C	03.26b (cycle 14)	3D	Cage is 'available' but cycle open; 02.26cK also moved through 1C
2B	02.26aK (cycle 10) and 05.26b (cycle 24)	G1.D and 2B	Two ongoing cycles in one cage